INGENIUM SEASON 14 - IDEA SUBMISSION
Team Name: 235rishavstar **Campus:** KIIT University

1. Project Name Project Rail-Pulse: On-Board Acoustic Interferometry for Predictive Rail Health Monitoring

2. Project Category Rail

3. Outline the real-life engineering problem that your project intends to solve? Why is the problem significant?

Railway infrastructure globally suffers from "Rolling Contact Fatigue" (RCF)—microscopic internal fractures in steel rails caused by repeated stress. Current inspection methods are inefficient: "Ultrasonic Inspection Cars" are expensive and infrequent (running only once every few months), while manual visual inspections can only detect surface-level damage. They cannot see inside the rail where dangerous fatigue cracks begin.

This problem is significant because undetected internal fractures lead to sudden rail snaps under load, causing catastrophic derailments. These accidents result in loss of life, severe environmental damage (in the case of freight), and billions of dollars in operational downtime. Furthermore, the current "schedule-based maintenance" approach is wasteful, as operators often replace healthy tracks too early just to be safe. There is a critical need for a continuous, real-time monitoring system that can detect internal structural anomalies during regular train operations, without requiring track shutdowns or specialized inspection vehicles.

4. **Please provide a description of the technology, tools or method used in building the solution.**

Our solution utilizes Piezoelectric Accelerometers mounted on the unsprung mass of the train bogie to capture high-frequency vibration data (up to 20kHz). We employ Fast Fourier Transform (FFT) signal processing to analyze the acoustic signature of the wheel-rail interaction. The data is processed locally using Edge AI (TinyML) on an NVIDIA Jetson Nano or Raspberry Pi. We use a Convolutional Neural Network (CNN) to classify vibration anomalies and transmit alerts via LoRaWAN/GSM modules to a cloud-based dashboard built on React and Firebase.

5. **Provide an abstract of the proposed solution, and how will it help in solving the above problem?**

Project Rail-Pulse proposes a shift from reactive to predictive maintenance by turning every operational train into a diagnostic tool. The core principle is Acoustic Fingerprinting. A healthy rail track produces a consistent resonant frequency and vibration pattern when a train wheel passes over it. However, a track with an internal hairline fracture or structural fatigue possesses different density and stiffness, resulting in a distinct variation in its harmonic response.

The system consists of a ruggedized sensor unit attached to the train bogie. As the train moves, the sensors capture raw vibration data. To handle the mechanical noise generated by the train itself, we implement an **Adaptive Noise Cancellation (ANC)** algorithm that filters out the engine and ambient noise, isolating the specific frequency generated by the wheel-rail contact point.

This purified signal is fed into a lightweight Machine Learning model trained on datasets of both healthy and fractured steel acoustic profiles. The processing happens locally on the device (Edge Computing) to reduce latency. When the AI detects a frequency shift consistent with a fracture (a "High-Energy Event"), it logs the severity and the precise GPS coordinates.

This data is visualized on a "Digital Twin" dashboard for railway engineers. Instead of walking miles of track, maintenance teams are directed to the exact meter where a stress fracture is forming. This solution solves the problem by enabling continuous, daily monitoring of the entire rail network without disrupting schedules. It detects internal flaws before they reach the surface, preventing derailments and optimizing maintenance budgets.

6. Please highlight the top 3 expected benefits of the Product / Solution for the end user. Who are your primary beneficiaries?

i. **Zero-Downtime Inspection:** Unlike traditional inspection cars that block traffic, Rail-Pulse operates on regular passenger/freight trains. This allows for continuous monitoring of track health without disrupting train schedules or reducing network capacity.
ii. **Internal Defect Detection:** While camera-based systems only see surface cracks, our acoustic approach detects changes in the internal density of the steel. This identifies dangerous fatigue cracks inside the rail long before they become visible or critical.
iii. **Cost-Efficient Maintenance:** By pinpointing exact defect locations, operators can switch to "predictive maintenance"—repairing only what is actually damaged. This saves millions in unnecessary material costs and manual labor.

Primary Beneficiaries: Railway Operators (e.g., Indian Railways), Freight & Logistics Companies, and the Public (Commuter Safety).

7. How much time have you and your team members spent in building the solution (in hours)? What roles do different team members play? Total Time: Approx. 60 Hours (Research & Architecture Design). Roles:

Sole Developer (Myself): I am responsible for the entire full-stack development, including the mechanical feasibility study, sensor selection, Edge AI model architecture, and the cloud dashboard design.

8. **Can your idea be improved or scaled further? What are the next steps if given an opportunity to scale it further?**

Yes, the system is highly scalable. The immediate next step is Federated Learning, allowing multiple trains to share data and improve the global AI model's accuracy across different terrains and weather conditions. We also plan to integrate Energy Harvesting (using the track vibrations to power the sensors via piezo-electric generators), making the device self-sustaining. Finally, we can expand the algorithm to detect "Wheel Flats" (damage to the train wheels themselves), offering a dual-purpose safety solution.

9. **Do you have a working model or prototype available? If yes, briefly describe the current stage of its development.**

Currently, the project is in the Design & Simulation Stage. I have validated the feasibility of the concept using software simulations and have completed the architectural design for the sensor node and the AI pipeline. I have also designed the UI mockups for the dashboard. The next phase is hardware procurement and physical integration.